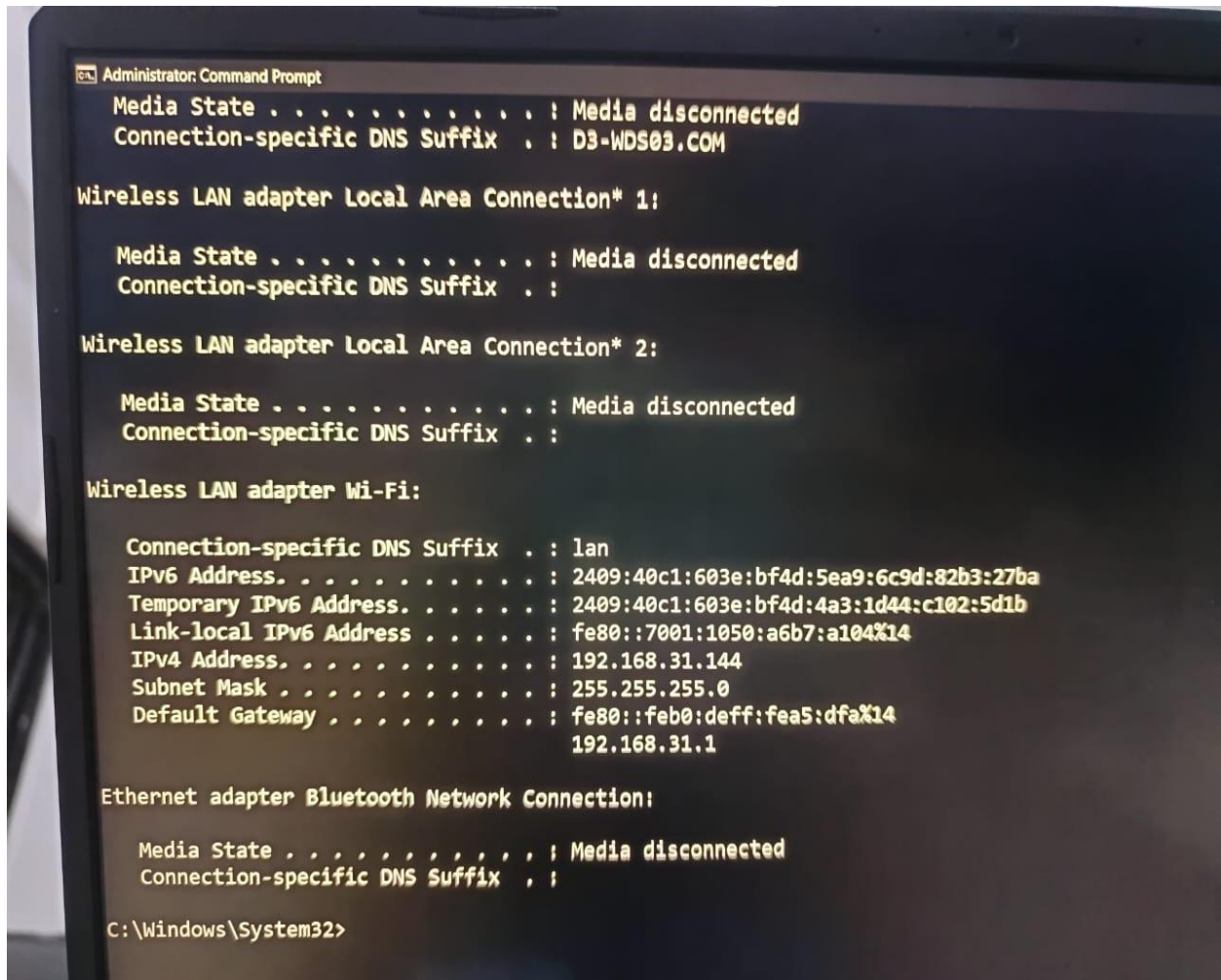


CONFIGURE IPv6 ADDRESSING

1. Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).



```
Administrator: Command Prompt
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . : D3-WDS03.COM

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : lan
IPv6 Address. . . . . : 2409:40c1:603e:bf4d:5ea9:6c9d:82b3:27ba
Temporary IPv6 Address. . . . . : 2409:40c1:603e:bf4d:4a3:1d44:c102:5d1b
Link-local IPv6 Address . . . . . : fe80::7001:1050:a6b7:a104%14
IPv4 Address. . . . . : 192.168.31.144
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::feb0:deff:fea5:dfa%14
                             192.168.31.1

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :

C:\Windows\System32>
```

2. Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

Hint: Use the ping6 or ping -6 command for IPv6 testing.

```
Address Type      : Other
Skip as Source    : false

C:\Windows\System32>ping -6 fd00:1:1::100

Pinging fd00:1:1::100 with 32 bytes of data:
Reply from fd00:1:1::100: time<1ms
Reply from fd00:1:1::100: time<1ms
Reply from fd00:1:1::100: time<1ms
Reply from fd00:1:1::100: time<1ms

Ping statistics for fd00:1:1::100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Windows\System32>
```

3. Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

Constraint: Document the steps and the observed address behavior in a text file.

OLD ADDRESS:-

Temporary IPv6 Address. : 2409:40c1:603e:bf4d:21a3:aa21:9745:9062

NEW ADDRESS:-

Temporary IPv6 Address. : 2409:40c1:603e:bf4d:bde9:a1a2:87ab:c368

4. Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

```
# Restaurant names and IPv6 addresses

restaurants = {
    "Dominos Pizza": "fd00:1:1::101",
    "Pizza Hut": "fd00:1:1::102",
    "McDonald's": "fd00:1:1::103",
    "Burger King": "fd00:1:1::104",
    "Subway": "fd00:1:1::105"
}

# User input
search = input("Enter restaurant name: ").lower()

# Search for matching restaurant
found = False

for name, ipv6 in restaurants.items():
    if search in name.lower():
        print("Restaurant:", name)
        print("IPv6 Address:", ipv6)
        found = True

if not found:
    print("Restaurant not found.")
```

5. Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

IPv6 Subnetting Scenario and Configuration

AI-Generated IPv6 Subnetting Scenario

Organization prefix: fd00:1234:abcd::/48

Three subnets:

1. Admin Network
 - Subnet: fd00:1234:abcd:1::/64
2. Staff Network
 - Subnet: fd00:1234:abcd:2::/64
3. Guest Wi-Fi
 - Subnet: fd00:1234:abcd:3::/64

Assigned Address

I selected the Staff Network.

IPv6 Address: fd00:1234:abcd:2::100/64

Configuration Steps (Windows)

1. Open Command Prompt as Administrator.
2. Check the adapter name: netsh interface ipv6 show interfaces
3. Assign the IPv6 address: netsh interface ipv6 set address "Wi-Fi"
fd00:1234:abcd:2::100
4. Verify: ipconfig